Adaptive Graph Modeling With Self-Training for Heterogeneous Cross-Scene Hyperspectral Image Classification

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***Abstract*— The small-sample-size problem of hyperspectral image (HSI) classification has recently gained considerable attention. Cross-scene HSI classification has emerged as an effective solution to this problem. In real-world applications, different HSI scenes are often captured by diverse sensors, resulting in variations between scenes. Graph modeling, as a method to represent relationships, leverages semantic information to establish connections between scenes, thereby facilitating transfer learning by aligning their features. However, in scenarios with only a few labeled target samples, the resulting graph is typically sparse and can only capture weak cross-scene relationships. Studies have shown that a dense and fault-tolerant graph is beneficial for transfer learning in small-sample-size cases. Consequently, we propose a novel heterogeneous transfer learning approach called adaptive graph modeling with self-training (AGM-ST). Unlike conventional graph modeling methods that employ predefined graph weights, adaptive graph modeling (AGM) employs a learnable network to generate graph weights based on the similarities of spectral–spatial features. Additionally, an adaptive cutoff threshold is trained to eliminate weak relationships between samples that may be potentially incorrect. Subsequently, a cross-scene graph loss is designed based on the generated graph to align the feature spaces of the source and target scenes. Furthermore, the unlabeled samples from the target scene are gradually updated with pseudo labels using the self-training (ST) technique, which enhances semantic information and improves graph modeling. Experimental evaluations conducted on three cross-scene HSI datasets have demonstrated the effectiveness of the proposed AGM-ST approach.**

***Index Terms*— Adaptive graph modeling (AGM), cross-scene classification, heterogeneous transfer learning, hyperspectral image (HSI), self-training (ST).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spatial and spectral information, making them valuable for accurate material identification and land-cover classification. In HSI applications, pixel classification has emerged as a prominent research area, garnering significant attention. However, due to the limited availability of labeled samples in HSIs [1], [2], [3], pixel classification often suffers from the challenge of small sample sizes [4], [5], [6]. Consequently, conventional classification methods tend to exhibit overfitting and yield subpar accuracy. Numerous methods have been proposed to address these issues, with cross-scene HSI classification being a popular and effective approach [7].

The cross-scene HSI classification problem assumes a scene with abundant samples (referred to as the source scene) and a scene with limited samples (known as the target scene). The primary objective of cross-scene HSI classification is to enhance the classification accuracy of the target scene by leveraging the knowledge from the source scene, thus addressing the transfer learning or domain adaptation problem [8], [9]. Drawing inspiration from traditional machine learning, numerous methods have applied transfer learning to HSI classification, yielding commendable performance [10], [11], [12]. In [13], the proposed method achieves transfer learning through rotation-based ensemble learning and transfer component analysis (TCA). In [14], a two-stage deep domain adaptation method is developed by combining maximum mean discrepancy (MMD) and pairwise loss. In [15], two projection matrices are utilized to map the source and target scene samples into a shared subspace, facilitating transfer learning. In [16], a deep conditional distribution adaptive network establishes constraints between the source and target scenes. In [17], adversarial learning is applied to attention-based spectral–spatial features for knowledge transfer between HSIs. In [18], a spatial–spectral weighted adversarial domain adaptation (SSWADA) network is proposed for cross-scene wetland mapping using HSIs. In SSWADA, adversarial learning is employed, incorporating a weighted discriminator to assign importance to source instances. In our previous research, we propose a transfer learning method based on multitask nonnegative matrix factorization (MTNMF), which demonstrates promising results [19].

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However, these methods are only applicable when the same sensor captures the source and target scenes. In real-world applications, the source and target scenes are often imaged by different sensors, giving rise to a heterogeneous transfer learning problem. This heterogeneity poses additional challenges that cannot be addressed by the aforementioned methods. To tackle this issue, several practical methods have been proposed for heterogeneous transfer learning. In [20], a mapping layer is designed to handle diverse feature inputs. In [8], a spectral feature adaptation module is combined with principal component analysis (PCA) to achieve cross-scene feature alignment. This approach successfully integrates cross-scene multiscale spectral–spatial information, yielding favorable results in heterogeneous transfer learning of HSIs. In [21], an intra-domain distribution extraction block (IDE-block) is designed to capture and aggregate intra-domain nonlocal relationships and inter-domain features. A cross-domain similarity aware block (CSA-block) is also developed to capture distribution similarities between domains. In [22], a 3-D separable ResNet network (3-D-SRNet) is proposed. It achieves promising outcomes by employing spatial and spectral separable 3-D convolutions in ResNet and unifying the dimensions of the two scenes through an adaptive pooling layer. In [23], to address the challenges posed by small sample sizes, the authors employ manifold augmentation to expand labeled data and learn features invariant to various scales and orientations of scene images using self-supervised contrastive learning. To overcome the few-shot learning (FSL) problem under domain shift, Li et al. [24] propose a deep cross-domain FSL (DCFSL) method. DCFSL utilizes a conditional adversarial domain adaptation strategy to tackle the domain shift problem. Moreover, by jointly utilizing the source and target scenes with FSL, DCFSL effectively transfers knowledge from the source scene and establishes a discriminative embedding model for the target scene.

Graph modeling is a promising approach for correlating cross-scene samples using semantic information. In [25], the adaptive regularization-based transfer learning (ARTL) algorithm is proposed. ARTL preserves manifold consistency between the source and target domains through graph regularization, enabling unified domain modeling. In [26], the dual-dictionary nonnegative matrix factorization (DDNMF) method performs feature alignment using graph regularization, maintaining relationships within a single scene and across different scenes. In [27], a dual-graph regularized transfer sparse coding method is introduced, establishing two distinct graphs based on inter-domain and intra-domain correlations. These graphs effectively transfer source domain knowledge to the target domain. In [28], an end-to-end heterogeneous deep transfer learning algorithm called cross-domain variational autoencoder (CDVAE) is proposed. CDVAE aligns two scenes using cross-domain discriminant graph modeling and achieves favorable results. In [29], a novel deep network called adaptive graph structure integration (DNAGSI) is presented. DNAGSI dynamically learns a graph structure of HSI and enhances its discriminative ability through a deeper network architecture. In [30], a weighted feature fusion of convolutional neural network and graph attention network (WFCG) is proposed. WFCG effectively combines the characteristics of a superpixel-based graph attention network and a pixel-based convolutional neural network (CNN), leading to improved HSI classification performance.

However, numerous transfer learning methods rely on a substantial number of labeled training samples in both scenes to establish robust correlations. In graph-based approaches, when the graph is constructed using insufficient labeled samples, it becomes too sparse to facilitate effective knowledge transfer between the source and target scenes. For instance, in [25], it is demonstrated that an extremely sparse graph captures only limited similarity information between samples, rendering it ineffective for transferring knowledge from the source scene. Furthermore, in [31], experimental results confirm that insufficient labeled samples can even lead to negative transfer.

A practical method for addressing the aforementioned challenges is self-training (ST), which involves generating pseudo labels to enhance the utilization of unlabeled samples [32], [33], [34]. However, it is important to note that these generated pseudo labels are often noisy, and relying solely on ST can lead to overconfident label beliefs placed in incorrect classes, resulting in distorted embeddings [35]. Additionally, Zhang and Zhang [36] emphasize that incorrect pseudo labels for target domain data can significantly deteriorate the results. Therefore, using pseudo-labels alone to tackle small-sample-size problems often proves challenging in achieving the desired effect. To mitigate the impact of incorrect pseudo labels, we propose a novel approach called adaptive graph modeling with ST (AGM-ST), which combines ST with adaptive graph modeling (AGM). AGM constructs soft relationships based on sample similarities, addressing the limitations of noisy pseudo labels. In this work, AGM adaptively generates graph weights between samples using a learnable network. Based on the generated graph, a cross-scene graph loss is then employed to strengthen the correlation between cross-scene samples and enhance classification accuracy. Finally, pseudo labels are generated for unlabeled samples in the target scene through the ST process, augmenting the semantic information of these samples. The main contributions of this method are as follows.

1) Unlike conventional graph modeling approaches that rely on predefined graph weights, AGM adaptively models soft relationships between samples using a learnable network. It establishes a more flexible graph connecting the source and target scenes. Additionally, a learnable cutoff threshold within AGM helps eliminate potentially incorrect relationships, resulting in a more robust graph for cross-scene transfer learning.
2) Given the small sample size in the target scene, the initial graph represents weak relationships between the source and target scenes. The iterative ST procedure increases labeled samples in the target scene using the pseudo labeling technique. These augmented pseudo labels contribute to constructing a stronger cross-scene graph, thereby establishing more robust source-to-target relationships.

The following parts of this article are organized as follows. Section II provides a brief overview of related works

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in the areas of transfer learning, graph modeling, and ST. Section III describes the proposed heterogeneous transfer learning algorithm, AGM-ST. Section IV presents the experimental results obtained from applying AGM-ST to cross-scene HSI datasets. Finally, Section V concludes the article by summarizing the findings and contributions of the proposed AGM-ST algorithm, and proposes future research ideas.

II. Related Works

This work combines AGM and ST as its fundamental modules. Therefore, this section provides a brief review of these essential components.

A. Transfer Learning and Graph Modeling

Transfer learning is widely recognized as an effective approach to address small-sample-size problems, particularly in overcoming the disparities between the source and target domains [37], [38], [39]. There are four main categories of transfer learning: instance-based transfer learning, network-based transfer learning, adversarial-based transfer learning, and mapping-based transfer learning [40].

This work focuses on the graph modeling method for mapping-based transfer learning. Graph modeling has proven to be valuable in various tasks as it captures the correlations between samples. For instance, in [41], the Eigenmaps method utilizing the graph Laplacian is proposed for dimension reduction and data representation. Similarly, in [42], a graph-regularized nonnegative matrix factorization (GNMF) model is introduced, establishing a graph to preserve the manifold structure of similar samples in the feature matrix. Additionally, in [43], nonnegative discriminative manifold learning based on a discriminative graph is employed for feature representation.

Cross-domain graph modeling often requires labeled samples from both the source and target domains due to the differences in feature distributions or feature spaces. However, when the target domain has only a limited number of labeled samples, it leads to poor performance in graph modeling, resulting in weak associations between samples [25] and even negative transfer effects [31]. To address this issue, we propose AGM-ST, which generates adaptive graph weights between unlabeled samples, allowing for more effective cross-domain graph modeling.

B. ST Algorithms

In transfer learning tasks, the target domain often suffers from a scarcity of labeled samples. In such cases, the generation of pseudo labels through ST has emerged as a promising alternative for transfer learning [35], [44]. A typical ST method consists of several steps: first, a classification model is trained using the available labeled data. Next, this trained model generates pseudo labels for the unlabeled samples, thereby expanding the label information. Finally, the classification model is re-trained by jointly optimizing the loss function using both the true and pseudo labels [32]. ST has made significant progress in deep learning, as demonstrated in various works. For instance, in [45], a class-balanced ST method is proposed, defining ST as a unified loss minimized by an end-to-end architecture, yielding notable results in unsupervised domain adaptation. Similarly, in [44], pseudo labels in the target domain are iteratively updated, enabling joint optimization of class-wise structure preservation and classifier adaptation.

Inspired by effectiveness of ST, pseudo labels can help establish a discriminative graph between scenes. However, it is essential to note that the predicted pseudo labels are not always entirely accurate. Relying solely on these generated pseudo labels as ground-truth can lead to overconfident mistakes, propagated errors, and negative transfer effects [35]. Therefore, in this work, we propose to create graph weights based on the similarities between samples, representing soft relationships that facilitate improved transfer learning. Meanwhile, ST is employed to generate pseudo labels for classification, leveraging their benefits while mitigating the potential issues associated with relying solely on noisy pseudo labels.

III. Proposed AGM-ST

This section introduces our novel method for heterogeneous transfer learning called AGM-ST. The AGM component utilizes a learnable network to generate graph weights adaptively, and then aligns the cross-scene features with a specially designed graph loss, enhancing the correlation between cross-scene samples. On the other hand, the ST component is employed to generate pseudo labels, effectively utilizing the unlabeled samples in the target scene. We provide detailed descriptions of this method in Sections III-A–III-C. Additionally, we summarize the symbols frequently used throughout this article in Table I.

A. Adaptive Graph Modeling

In this work, we divide the AGM component into two stages for graph weight generation and classification, respectively. Both stages share the same architecture and train the network weights successively. Here is a specific description of these two stages.

1) First Stage of AGM: Section II-A discussed how existing graph modeling methods can suffer from weak transferability or negative transfer [25], [31]. To address these issues, the first stage of AGM in our proposed AGM-ST method trains a network to generate adaptive graph weights, enhancing the robustness of graph modeling. The graph construction mainly consists of the following two steps.

1) Training a feature extraction network using abundant labeled samples in the source scene and limited labeled samples in the target scene, with the goal of minimizing the distribution discrepancy between the two scenes.
2) Calculating graph weights based on the similarities between samples, which are determined using the extracted features.

To improve the performance of the algorithm, it is essential to have a robust feature extraction method. In the context of HSI classification, spectral–spatial features have demonstrated excellent performance [8], [46], [47]. In this work, we design a spectral–spatial feature block (SSFB) to capture both spectral and spatial information, as shown in Fig. 1. The SSFB

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TABLE I
SYMBOLS AND DESCRIPTIONS RELATED TO THE PROPOSED ALGORITHM IN THIS ARTICLE

Symbol	Description	Type
$\mathcal{S}$	Source scene	-
$\mathcal{T}$	Target scene	-
$N^{\mathcal{S}}$	Number of samples in the source scene	scalar
$N^{\mathcal{T}}$	Number of samples in the target scene	scalar
$M^{\mathcal{S}}$	Input feature dimension in the source scene	scalar
$M^{\mathcal{T}}$	Input feature dimension in the target scene	scalar
$\mathbf{x}$	Input feature vectors	vector
$\mathbf{X}$	Input matrix containing all feature vectors	matrix
$\mathbf{z}$	Hidden feature vector produced by the hidden layer	vector
$\mathbf{Z}$	Matrix containing all hidden vectors	matrix
r	Subspace dimensions of spectral and spatial branches	scalar
$a_{i,j}^{\mathcal{ST}}{}_{\text{intra}}$	Indicator of intra-class samples	scalar
$a_{k,l}^{\mathcal{ST}}{}_{\text{inter}}$	Indicator of inter-class samples	scalar
U	Normalization factor of graph weights	scalar
$\mathbf{W}^{\mathcal{ST}}$	Adjacency matrix of the graph in the second stage of AGM	matrix
$w_{i,j}^{\mathcal{ST}}$	An element of the adjacency matrix $\mathbf{W}^{\mathcal{ST}}$	scalar
ζ	Balancing factor between VAE losses over the source and target scenes	scalar
β	Balancing factor between intra-class and inter-class graph losses	scalar
θ	Weight of cross-entropy loss	scalar
λ	Weight of graph loss	scalar
η	Weight of balancing loss	scalar
τ	Adaptive cutoff threshold	scalar
$\mathcal{L}_{\text{VAE}}$	VAE loss	scalar
$\mathcal{L}_{\text{DD}}$	Distribution discrepancy loss	scalar
$\mathcal{L}_{\text{B}}$	Balancing loss	scalar
$\mathcal{L}_{\text{CE}}$	Cross-entropy loss	scalar
$\mathcal{L}_{\text{G}}$	Graph loss	scalar
$\mathcal{L}_{\text{first}}$	Loss function for the first stage of AGM	scalar
$\mathcal{L}_{\text{second}}$	Loss function for the second stage of AGM	scalar

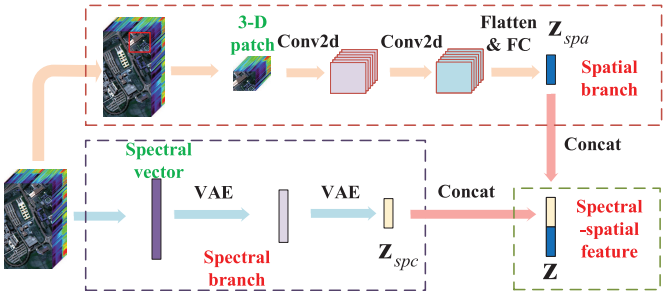


Fig. 1. Architecture of the proposed SSFB.

consists of two branches, where spectral branch consists of two variational autoencoder (VAE) layers [28], [48], while spatial branch includes two 2-D convolutional layers, a flatten layer and a fully connected (FC) layer. Finally, spectral and spatial branches are merged by concatenating spectral features $\mathbf{z}_{\text{spe}}$ and spatial features $\mathbf{z}_{\text{spa}}$

$$\mathbf{z} = \text{concat}(\mathbf{z}_{\text{spe}}, \mathbf{z}_{\text{spa}}). \tag{1}$$

In this work, we ensure that the spectral feature $\mathbf{z}_{\text{spe}}$ and the spatial feature $\mathbf{z}_{\text{spa}}$ have the same dimension r. As a result, the concatenated spectral–spatial feature is represented as $\mathbf{z} \in \mathbb{R}^{2r}$. To represent the spectral–spatial features in the source scene, we use $\mathbf{Z}^{\mathcal{S}} \in \mathbb{R}^{N^{\mathcal{S}} \times 2r}$, where $N^{\mathcal{S}}$ denotes the number of samples in the source scene. Similarly, for the target scene,

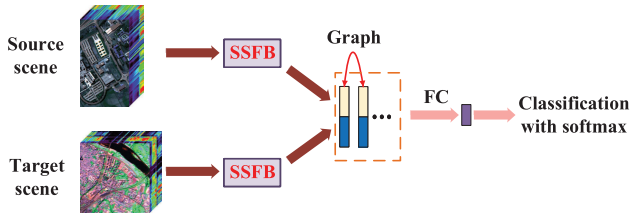


Fig. 2. Architecture of the proposed AGM.

we use $\mathbf{Z}^{\mathcal{T}} \in \mathbb{R}^{N^{\mathcal{T}} \times 2r}$, where $N^{\mathcal{T}}$ represents the number of samples in the target scene. These representations capture the combined spectral and spatial information for each sample in their respective scenes.

As depicted in Fig. 2, AGM utilizes two SSFBs for feature extraction in both the source and target scenes. The SSFBs extract spectral–spatial features with the same dimension, enabling them to be shared in the subspace. These features, the same dimension, are then inputted into a shared network, which typically includes an FC layer and a classification layer, for the classification task. To achieve the graph weight generation task in the first stage of AGM, a specially designed loss function is employed, consisting of four parts: 1) VAE loss $\mathcal{L}_{\text{VAE}}$; 2) distribution discrepancy loss $\mathcal{L}_{\text{DD}}$; 3) balancing loss $\mathcal{L}_{\text{B}}$; and 4) cross-entropy loss $\mathcal{L}_{\text{CE}}$.

a) VAE loss: The first part is the VAE loss for spectral feature extraction, which extends the conventional VAE to a cross-scene model incorporating multiple VAE layers [28]. We break down the components of the VAE loss for detailed analysis.

1) *Basic VAE Loss:* For a basic VAE layer, the loss function $\mathcal{L}_{\text{VAE}}^{\text{basic}}$ consists of two components: the reconstruction loss $\mathcal{L}_{\text{REC}}$ and the Kullback–Leibler (KL) divergence loss $\mathcal{L}_{\text{KL}}$ [48]

$$\mathcal{L}_{\text{VAE}}^{\text{basic}} = \mathcal{L}_{\text{REC}} + \mathcal{L}_{\text{KL}}. \tag{2}$$

The reconstruction loss $\mathcal{L}_{\text{REC}}$ measures the discrepancy between the original input data and the reconstructed output

$$\mathcal{L}_{\text{REC}} = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z})] \tag{3}$$

where $q(\mathbf{z}|\mathbf{x})$ represents the latent distribution given the input data $\mathbf{x}$, and $p(\mathbf{x}|\mathbf{z})$ represents the decoder's ability to reconstruct the input data from the latent space. KL divergence loss $\mathcal{L}_{\text{KL}}$ encourages the latent space $q(\mathbf{z}|\mathbf{x})$ to approximate a Gaussian distribution $p(\mathbf{z})$. The KL divergence measures the difference between these two distributions

$$\mathcal{L}_{\text{KL}} = \text{KL}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z})). \tag{4}$$

2) *Cross-Domain VAE Loss for Each Layer:* When extended to a cross-scene model, each layer has a separate VAE loss for both the source and target scenes. The cross-domain VAE loss for the ith layer is defined as

$$\mathcal{L}_{\text{VAE}}^{(i)} = \mathcal{L}_{\text{VAE}}^{\mathcal{S}(i)} + \zeta \mathcal{L}_{\text{VAE}}^{\mathcal{T}(i)} \tag{5}$$

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where $\mathcal{L}_{\text{VAE}}^{\mathcal{S}(i)}$ and $\mathcal{L}_{\text{VAE}}^{\mathcal{T}(i)}$ are the VAE losses of the source and target scenes, respectively. $\zeta = \|\mathbf{X}^{\mathcal{S}}\|_F^2 / \|\mathbf{X}^{\mathcal{T}}\|_F^2$ is the balancing factor [31], [49]. $\mathbf{X}^{\mathcal{S}} \in \mathbb{R}^{N^{\mathcal{S}} \times M^{\mathcal{S}}}$ and $\mathbf{X}^{\mathcal{T}} \in \mathbb{R}^{N^{\mathcal{T}} \times M^{\mathcal{T}}}$ mean the input matrices containing all feature vectors in the source and target scenes, where $M^{\mathcal{S}}$ and $M^{\mathcal{T}}$ are the numbers of input feature dimensions, and $N^{\mathcal{S}}$ and $N^{\mathcal{T}}$ are the number of samples. $\|\cdot\|_F$ is the Frobenius norm.

3) *Final VAE Loss:* It is composed of the sum of the VAE losses for each layer

$$\mathcal{L}_{\text{VAE}} = \sum_i \mathcal{L}_{\text{VAE}}^{(i)}. \tag{6}$$

In this work, we adopt two VAE layers to achieve spectral feature extraction.

b) Distribution discrepancy loss: The second part is distribution discrepancy loss, which reflects the class co-occurrence semantics between source and target scenes. It reduces the distance of cross-scene intra-class samples while pushing away cross-scene inter-class samples, thus reducing the distribution discrepancy

$$\mathcal{L}_{\text{DD}} = \sum_{i,j} a_{i,j\ \text{intra}}^{\mathcal{ST}} \left\|\mathbf{Z}_{i,\cdot}^{\mathcal{S}} - \mathbf{Z}_{j,\cdot}^{\mathcal{T}}\right\|_2^2 - \beta \sum_{k,l} a_{k,l\ \text{inter}}^{\mathcal{ST}} \left\|\mathbf{Z}_{k,\cdot}^{\mathcal{S}} - \mathbf{Z}_{l,\cdot}^{\mathcal{T}}\right\|_2^2. \tag{7}$$

The first term of (7) enforces the intra-class samples to be closer to each other, where

$$a_{i,j\ \text{intra}}^{\mathcal{ST}} = \begin{cases} \dfrac{1}{U_{\text{intra}}^{\mathcal{ST}}}, & \text{class}\left(\mathbf{X}_{i,\cdot}^{\mathcal{S}}\right) = \text{class}\left(\mathbf{X}_{j,\cdot}^{\mathcal{T}}\right) \\ 0, & \text{class}\left(\mathbf{X}_{i,\cdot}^{\mathcal{S}}\right) \neq \text{class}\left(\mathbf{X}_{j,\cdot}^{\mathcal{T}}\right) \end{cases} \tag{8}$$

is the indicator of intra-class samples, and $U_{\text{intra}}^{\mathcal{ST}} = \sum_{i,j} a_{i,j\ \text{intra}}^{\mathcal{ST}}$ is the normalization factor. The second term of (7) is similarly defined to enforce the inter-class samples to be farther away from each other, where

$$a_{k,l\ \text{inter}}^{\mathcal{ST}} = \begin{cases} \dfrac{1}{U_{\text{inter}}^{\mathcal{ST}}}, & \text{class}\left(\mathbf{X}_{k,\cdot}^{\mathcal{S}}\right) \neq \text{class}\left(\mathbf{X}_{l,\cdot}^{\mathcal{T}}\right) \\ 0, & \text{class}\left(\mathbf{X}_{k,\cdot}^{\mathcal{S}}\right) = \text{class}\left(\mathbf{X}_{l,\cdot}^{\mathcal{T}}\right) \end{cases} \tag{9}$$

is the indicator of inter-class samples, and $U_{\text{inter}}^{\mathcal{ST}} = \sum_{i,j} a_{i,j\ \text{inter}}^{\mathcal{ST}}$ is the normalization factor. $\beta > 0$ is the balancing factor of intra-class and inter-class samples.

c) Balancing loss: The third part is the balancing loss for balancing intra-class and inter-class correlations. Recalling the aforementioned distribution discrepancy loss, we can find the minimization of $\mathcal{L}_{\text{DD}}$ can be achieved when $\sum_{k,l} a_{k,l\ \text{inter}}^{\mathcal{ST}} \|\mathbf{Z}_{k,\cdot}^{\mathcal{S}} - \mathbf{Z}_{l,\cdot}^{\mathcal{T}}\|_2^2 \to +\infty$ (inter-class samples are infinitely far away), even if $\sum_{i,j} a_{i,j\ \text{intra}}^{\mathcal{ST}} \|\mathbf{Z}_{i,\cdot}^{\mathcal{S}} - \mathbf{Z}_{j,\cdot}^{\mathcal{T}}\|_2^2$ remains a large value (intra-class samples are not close to each other). It brings difficulty to optimization, and may lead to numerical problems. To address this problem, intra-class and inter-class correlations should be balanced. Hence, we propose a balancing loss. Assuming that the cosine similarity is adopted, the similarity between $\mathbf{Z}_{i,\cdot}^{\mathcal{S}}$ and $\mathbf{Z}_{j,\cdot}^{\mathcal{T}}$ is defined as

$$s_{i,j}^{\mathcal{ST}} = \cos\left(\mathbf{Z}_{i,\cdot}^{\mathcal{S}}, \mathbf{Z}_{j,\cdot}^{\mathcal{T}}\right). \tag{10}$$

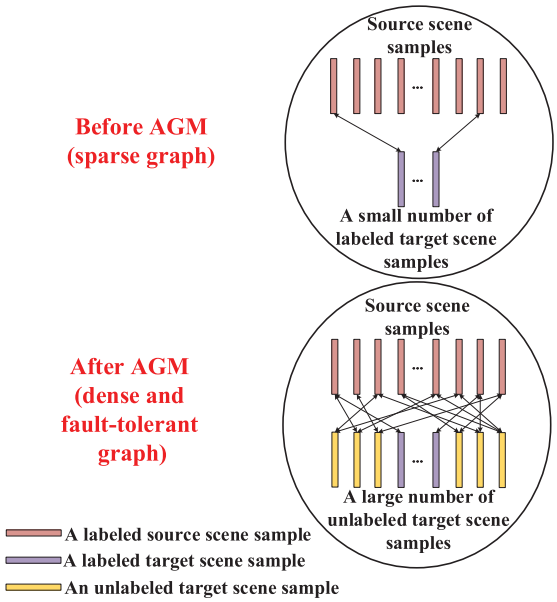


Fig. 3. Comparison of the graph before and after AGM.

Then the balancing loss is defined as

$$\mathcal{L}_{\text{B}} = \sum_{i,j} a_{i,j\ \text{intra}}^{\mathcal{ST}} \left(s_{i,j}^{\mathcal{ST}} - \tau\right)^2 + \sum_{k,l} a_{k,l\ \text{inter}}^{\mathcal{ST}} \left(s_{k,l}^{\mathcal{ST}} - \tau\right)^2. \tag{11}$$

Two terms are the summations over intra-class and inter-class samples across source and target scenes. In balancing loss, the learnable parameter τ is a trade-off between intra-class and inter-class correlations. After training, τ also plays an important role as the cutoff threshold to determine whether two samples are intra-class samples, i.e., in the subsequent stage of AGM, we cut off the correlations when $s_{i,j}^{\mathcal{ST}} < \tau$.

d) Cross-entropy loss: The fourth part is the cross-entropy loss

$$\mathcal{L}_{\text{CE}} = \mathcal{L}_{\text{CE}}^{\mathcal{S}} + \mathcal{L}_{\text{CE}}^{\mathcal{T}} \tag{12}$$

where $\mathcal{L}_{\text{CE}}^{\mathcal{S}}$ and $\mathcal{L}_{\text{CE}}^{\mathcal{T}}$ are the cross-entropy losses of the source and target scenes, respectively. It should be noted that $\mathcal{L}_{\text{CE}}^{\mathcal{S}}$ and $\mathcal{L}_{\text{CE}}^{\mathcal{T}}$ are separately averaged over all samples within each scene.

Combining these losses, the total loss function for the first stage of AGM can be defined as follows:

$$\mathcal{L}_{\text{first}} = \mathcal{L}_{\text{VAE}} + \lambda \mathcal{L}_{\text{DD}} + \eta \mathcal{L}_{\text{B}} + \theta \mathcal{L}_{\text{CE}} \tag{13}$$

where λ, η, and θ are the weights of corresponding losses.

As shown in Fig. 3, after the first stage of AGM, we can use the similarities of features between the source and target scenes to build an adaptive graph, whose adjacency matrix $\mathbf{W}^{\mathcal{ST}}$ is represented as

$$w_{i,j}^{\mathcal{ST}} = \max\left(s_{i,j}^{\mathcal{ST}} - \tau, 0\right). \tag{14}$$

It will be fed into the second stage of AGM for cross-domain transfer learning.

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TABLE II
NUMBER OF TRAINABLE PARAMETERS IN AGM

Layer	Source stream	Target stream
Spectral branch	• First VAE layer: $151M^{\mathcal{S}}+100$; • Second VAE layer: $152r+50$;	• First VAE layer: $151M^{\mathcal{T}}+100$; • Second VAE layer: $152r+50$;
Spatial branch	• First convolutional layer: $30M^{\mathcal{S}}+30$; • Second convolutional layer: 2710 ;	• First convolutional layer: $30M^{\mathcal{T}}+30$; • Second convolutional layer: 2710 ;
Fully connected layer (shared)	$2rc+c$	
Total	$181M^{\mathcal{S}}+181M^{\mathcal{T}}+304r+2rc+c+5780$	

2) Second Stage of AGM: With the adaptive graph calculated in the first stage, the cross-scene classification is completed in the second stage of AGM. The second stage shares the same architecture used in the first stage (see Fig. 2), but the loss function is specially designed due to a different goal. To maintain the cross-domain correlations, a graph loss $\mathcal{L}_{\mathrm{G}}$ is defined as follows:

$$\mathcal{L}_{\mathrm{G}}=\sum_{i,j} w_{i,j}^{\mathcal{ST}}\left\|\mathbf{Z}_{i,\cdot}^{\mathcal{S}}-\mathbf{Z}_{j,\cdot}^{\mathcal{T}}\right\|_2^2. \tag{15}$$

The graph loss implies that two cross-scene samples should be similar when they potentially belong to the same land-cover class. Apart from graph loss $\mathcal{L}_{\mathrm{G}}$, VAE loss $\mathcal{L}_{\mathrm{VAE}}$ is required for VAE-based spectral feature extraction, and cross-entropy loss $\mathcal{L}_{\mathrm{CE}}$ is introduced for the classification task. Finally, the loss for the second stage is represented as

$$\mathcal{L}_{\mathrm{second}}=\mathcal{L}_{\mathrm{VAE}}+\lambda\mathcal{L}_{\mathrm{G}}+\theta\mathcal{L}_{\mathrm{CE}} \tag{16}$$

where the weights λ and θ are set the same as the first stage.

Finally, the number of trainable parameters in AGM is analyzed in Table II, where c is the number of land-cover classes in the dataset. In addition to the training parameters listed in Table II, there is an additional training parameter τ in the first stage of AGM.

B. Self-Training

To better obtain label information for unlabeled samples and improve the robustness of the graph generated in the first stage of AGM, we employ ST to generate pseudo labels. Specifically, after each round of the AGM procedure, the graph modeling samples (unlabeled samples used to generate graph weights adaptively) are fed into the model for prediction. These samples are then ranked based on their softmax probabilities. To prevent noisy pseudo labels from negatively affecting the classification, we select the top n samples with high rankings to generate pseudo labels for the next round of training. Notably, these pseudo-labeled samples are not re-ranked by softmax probabilities in subsequent rounds. The ST process is applied after each round of AGM until all the pseudo labels are generated. This iterative approach helps incorporate additional labeled information from the model predictions, enhancing the AGM's ability to handle unlabeled samples and improving the overall performance and robustness of the model.

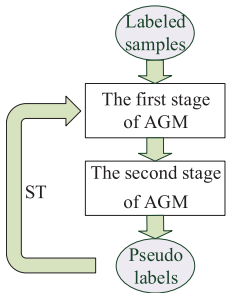


Fig. 4. Flowchart of the AGM-ST.

Algorithm 1 AGM-ST

Input:
Training samples from the source and target scenes, along with their corresponding labels.

Output:
Predicted labels of test samples.

1: **while** Any graph modeling sample has not been pseudo-labeled **do**
2: Optimize Eq. (13) to train the first stage of AGM.
3: Calculate the graph weights using Eq. (14).
4: Optimize Eq. (16) to train the second stage of AGM.
5: Perform predictions on the unlabeled graph modeling samples, sort their softmax probabilities, and assign pseudo labels to the samples with the highest rankings.
6: **end while**
7: Input the test samples for classification.

C. Overall Process of AGM-ST

With the aforementioned components, the overall process of AGM-ST is presented in Algorithm 1, and the corresponding flowchart is depicted in Fig. 4.

The specific stages of AGM-ST can be summarized as follows.

1) The first stage of AGM (lines 2–3 in Algorithm 1) trains a cross-scene feature extraction network based on class co-occurrence semantics. This learnable network generates adaptive graph weights, facilitating the integration of information between scenes.
2) The second stage of AGM (line 4 in Algorithm 1) focuses on cross-scene transfer learning. The specially designed graph loss connects the two scenes by leveraging the adaptive graph generated in the first stage of AGM. A joint cross-scene classification model is trained to enhance the transferability of learned features.

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TABLE III
NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN RPAVIAU-DPAVIAC DATASET

Class		Labeled samples	
No.	Name	Source scene (total/training)	Target scene (total/training)
1	Trees	3064/400	2424/5
2	Asphalt	6631/400	1704/5
3	Bitumen	1330/400	685/5
4	Shadow	947/400	241/5
5	Brick	3682/400	2237/5
6	Meadow	18649/400	1251/5
7	Soil	5029/400	1475/5
Total		39332/2800	10017/35

3) The ST procedure gradually assigns pseudo labels to unlabeled samples. By incorporating these pseudo-labeled samples, the AGM model is retrained, effectively leveraging additional labeled information and enhancing the model's robustness, especially in scenarios with limited labeled samples.

It is noteworthy that pseudo-labeled pixels obtained through ST will be utilized as training samples in subsequent rounds of ST. As a result, the graph will be updated to accurately depict the correlations between samples.

IV. EXPERIMENTS

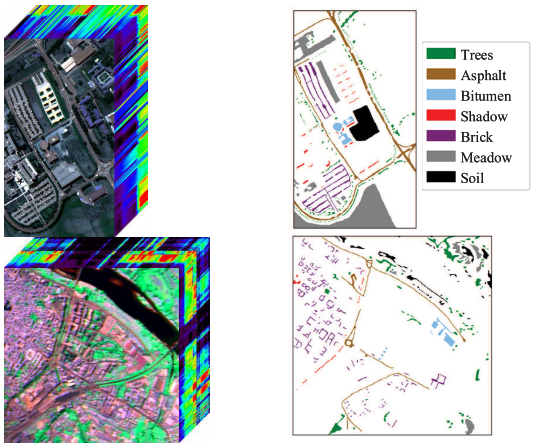
A. Datasets Description

In this section, we evaluate the performance of the proposed AGM-ST by conducting experiments using three heterogeneous cross-scene HSI datasets.

The first dataset, named ROSIS Pavia University (RPaviaU)-DAIS Pavia Center (DPaviaC), consists of two scenes captured by different sensors. The source scene, RPaviaU, was obtained using the ROSIS HSI sensor over the University of Pavia, Italy. It has dimensions of 610 × 340 × 103. The target scene, DPaviaC, was captured by the DAIS sensor over the center of Pavia city, Italy, and has dimensions of 400 × 400 × 72. These two scenes share seven land-cover classes, and the details of these classes can be found in Table III. The data cubes and ground-truth maps for the RPaviaU-DPaviaC dataset are shown in Fig. 5.

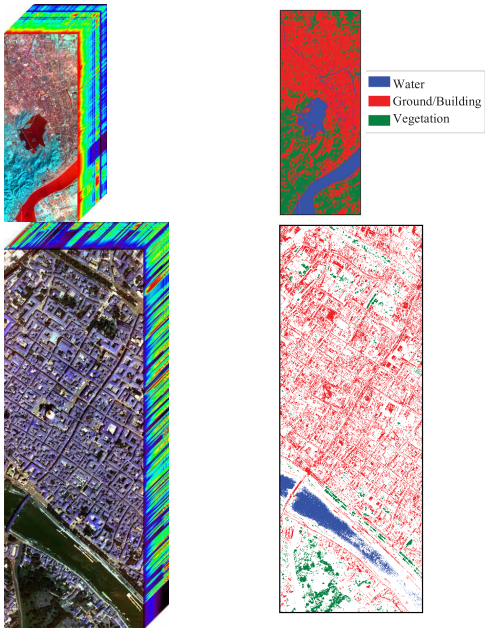
The second dataset, named EO-1 Hangzhou (EHangzhou)-ROSIS Pavia HR (RPaviaHR), also consists of two scenes. The source scene, EHangzhou, was captured by the EO-1 Hyperion sensor over Hangzhou city, Zhejiang, China, with dimensions of 590 × 230 × 198. The target scene, RPaviaHR, was captured by the ROSIS HSI sensor over Pavia city, Italy, with dimensions of 1400 × 512 × 102. These scenes share three land-cover classes, and the details of these classes can be found in Table IV. The data cubes and ground-truth maps for the EHangzhou-RPaviaHR dataset are shown in Fig. 6.

The third dataset, named Houston2018-Houston2013, also consists of two scenes. The source scene, Houston2018 [50], was captured over the University of Houston campus and its neighborhood, with dimensions of 601 × 2384 × 48. The target scene, Houston2013 [51], was captured over the University of Houston campus, with dimensions of 49 × 1905 × 144.



(a) (b)

Fig. 5. Data cubes and ground-truth maps of RPaviaU-DPaviaC. (a) Data cubes. (b) Ground-truth maps. (Top) source scene (RPaviaU), (Bottom) target scene (DPaviaC).



(a) (b)

Fig. 6. Source and target cubes and ground-truth maps of EHangzhou-RPaviaHR. (a) Data cubes. (b) Ground-truth maps. (Top) source scene (EHangzhou), (Bottom) target scene (RPaviaHR).

Houston2018 and Houston2013 were captured over different regions using different hyperspectral sensors, making Houston2018-Houston2013 a heterogeneous cross-scene HSI dataset. Seven common land-cover classes are adopted for cross-domain HSI classification, and the details of these classes can be found in Table V. The data cubes and ground-truth maps for the Houston2018-Houston2013 dataset are shown in Fig. 7.

B. Experimental Settings

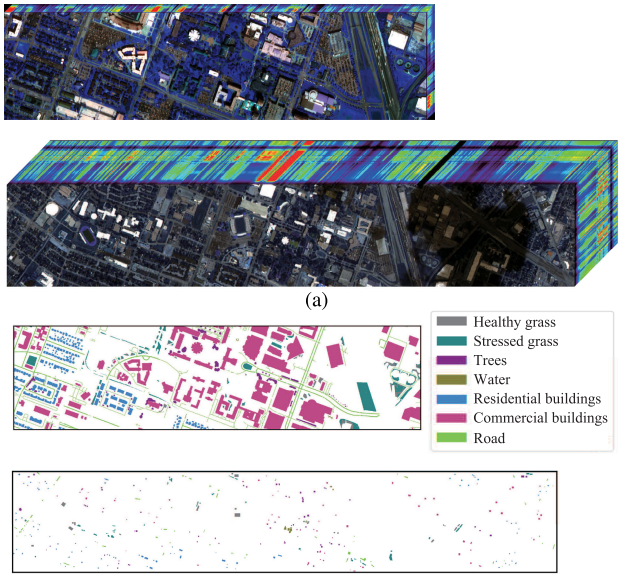
1) *Evaluation Metrics:* We evaluate the classification performance of each algorithm quantitatively in terms of three commonly used criteria: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ).

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TABLE IV

NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN EHANGZHOU-RPAVIAHR DATASET

Class		Labeled samples	
No.	Name	Source scene (total/training)	Target scene (total/training)
1	Water	18403/400	22525/5
2	Ground/Building	77450/400	145416/5
3	Vegetation	40207/400	22961/5
Total		136060/1200	190902/15



(a)

(b)

Fig. 7. Source and target cubes and ground-truth maps of Houston2018-Houston2013. (a) Data cubes. (b) Ground-truth maps. (Top) source scene (Houston2018), (Bottom) target scene (Houston2013).

TABLE V

NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN HOUSTON2018-HOUSTON2013 DATASET

Class		Labeled samples	
No.	Name	Source scene (total/training)	Target scene (total/training)
1	Healthy grass	9799/400	1251/5
2	Stressed grass	3250/400	1254/5
3	Trees	13595/400	1244/5
4	Water	266/266	325/5
5	Residential buildings	39772/400	1268/5
6	Commercial buildings	223752/400	1244/5
7	Road	45866/400	1252/5
Total		365552/2666	7838/35

Additionally, the classification accuracies are averaged over ten repeated experiments with random sampling.

2) *Compared Methods:* In order to evaluate the effectiveness of the proposed AGM-ST, the accuracies of the following ten different algorithms are analyzed and compared:

   a) *Support vector machine:* As the baseline algorithm, support vector machine (SVM) is used to classify the spectral features of HSI. In the SVM, we set the penalty parameter as $C \in \{10^{-3}, 10^{-2}, \ldots, 10^{2}\}$ and the parameter of the radial basis function (RBF) kernel as $\gamma \in \{1, 2, 4, 8, 16, 32\}$.

   b) *Hybrid spectral convolutional neural network:* Hybrid spectral convolutional neural network (HybridSN) is an efficient algorithm suitable for single-scene HSI classification [52]. HSI data is classified using a combination of 3-D and 2-D convolutional layers, which reduces computational complexity and achieves higher classification accuracy. In the HybridSN algorithm, we set the patch size of 3-D patches as 25×25, the feature dimension after PCA as 30, the optimizer as Adam (with a learning rate of 10^{-3}), and the number of training epochs as 10^3.

   c) *Target-domain spectral–spatial feature net:* Target-domain spectral-spatial feature net (TDSSFnet) is a single-scene algorithm proposed in our work. It utilizes only target scene samples during the training and testing processes. In TDSSFnet, target scene samples are fed into the SSFB (see Fig. 1) to generate spectral–spatial features. These features are then directly fed into the flatten layer, FC layer, and softmax layer for classification. In the TDSSFnet algorithm, we set the optimizer as Adam (with a learning rate of 10^{-3}) and the number of training epochs as 10^3.

   d) *3-D-SRNet:* 3-D-SRNet is an algorithm applied to cross-scene HSI classification [22]. It first uses the source scene samples for pre-training and then fine-tunes the network using the target scene samples, thereby transferring knowledge from the source scene to the target scene. The algorithm also introduces a 3-D residual unit to reduce network parameters and achieves good results in cross-scene HSI tasks. In the 3-D-SRNet algorithm, we set the patch size of 3-D patches as 9×9, the optimizer as Adam (with a learning rate of 10^{-3}), the number of pre-training epochs as 10^3, and the number of fine-tuning epochs as 10^3.

   e) *DDNMF:* DDNMF is a cross-scene classification algorithm [26]. It implements cross-domain feature embedding by applying intra-domain and inter-domain graph regularizations to the loss function of dictionary training, enabling transfer learning. The algorithm can achieve good results when sufficient labeled target scene samples are available. In the DDNMF algorithm, the graph regularization parameter is dynamically updated based on the number of iterations [31].

   f) *CDVAE:* In CDVAE, a two-stream VAE is proposed to extract features between two scenes, and a source–target discriminative graph is used to align their features [28]. CDVAE achieves both transferability and discriminability. In the CDVAE algorithm, we adopt the optimizer Adam with a learning rate of 10^{-3} and train for 10^3 epochs.

   g) *Graph information aggregation cross-domain FSL:* Graph information aggregation cross-domain FSL (Gia-CFSL) aims to address the challenges of heterogeneous cross-scene classification [21].

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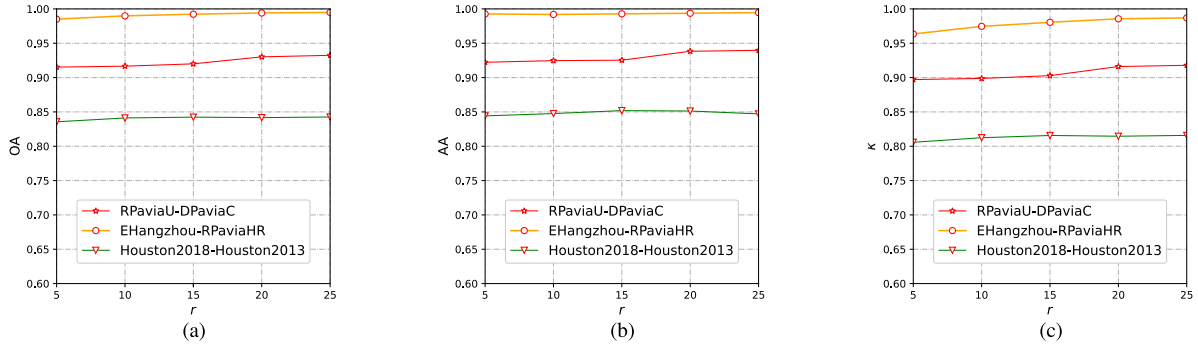


Fig. 8. Classification accuracies of AGM-ST with different r values. (a) OA. (b) AA. (c) κ.

In Gia-CFSL, an episodic learning pattern of FSL is employed to learn the class separability metric space within the source and target scenes. Additionally, IDE-block and CSA-block are adopted as domain alignment methods to aggregate intra-domain nonlocal relationships and enhance inter-domain similarities. In the Gia-CFSL algorithm, we set the patch size of 3-D patches as 9×9, the optimizer as Adam (with a learning rate of 10^{-3}), and the number of training epochs as 10^3.

h) *DCFSL:* DCFSL is a cross-domain FSL algorithm [24]. In DCFSL, a conditional adversarial domain adaptation strategy is utilized to overcome the domain shift, and FSL is performed in both the source and target scenes to effectively transfer knowledge from the source scene to the target scene while learning a discriminative embedding model for the target scene. In the DCFSL algorithm, we set the patch size of 3-D patches as 9×9, the optimizer as Adam (with a learning rate of 10^{-3}), and the number of training epochs as 10^3.

i) *AGM:* AGM is a fundamental model designed in this work, which adaptively generates graph weights between samples using a specially defined network and loss function. This adaptively generated graph is then effectively utilized to establish correlations between cross-scene samples and improve classification performance. In the AGM algorithm, we set the patch size of 3-D patches as 9×9. Various parameters of the loss function are chosen as follows: $\beta = 10^{-2}$, $\lambda = 10^{-5}$, $\eta = 10^3$, $\theta = 10^3$. The subspace dimension is selected from the set 5, 10, 15, 20, 25. The number of graph modeling samples in the target scene is set to 400. For the optimization process, the first stage uses Adam as the optimizer with a learning rate of 10^{-3} and weight decay parameter of 1, while the second stage also uses Adam with a learning rate of 10^{-3}. The training epochs for both stages are set to 10^3.

j) *AGM-ST:* AGM-ST is our proposed algorithm in this work. In addition to AGM's adaptive graph weight generation, ST generates pseudo labels between unlabeled target scene samples for classification. AGM-ST not only incorporates label information but also ensures a dense and fault-tolerant graph with a small number of labeled samples, effectively addressing the challenges of small-sample-size problems. In the AGM-ST algorithm, we set the patch size of 3-D patches as 9×9. Various parameters of the loss function are chosen as follows: $\beta = 10^{-2}$, $\lambda = 10^{-5}$, $\eta = 10^3$, $\theta = 10^3$. The subspace dimension is selected from the set 5, 10, 15, 20, 25. The number of graph modeling samples in the target scene is set to 400. For the optimization process, the first stage uses Adam as the optimizer with a learning rate of 10^{-3} and weight decay parameter of 1. The second stage also uses Adam with a learning rate of 10^{-3}. The training epochs for both stages are set to 10^3. In each round of ST, 100 pseudo labels are added.

3) *Sample Selection:* We simulate the most common situation in real-world cross-scene HSI classification, where the source scene contains a large number of labeled samples, while the target scene contains only a small number of labeled samples. In the single-scene experiments (including SVM, HybridSN, and TDSSFnet), only the target scene samples (five labeled samples of each class) are used for training. In the cross-scene experiments (including 3-D-SRNet, DDNMF, CDVAE, Gia-CFSL, DCFSL, AGM, and AGM-ST), both the source scene (400 labeled samples of each class) and the target scene (five labeled samples of each class) samples are fed into the network for training. It is worth noting that the Water class in the source scene of the Houston2018-Houston2013 dataset has fewer than 400 samples. Therefore, we select all samples from this class as training samples. For the special ST procedure in AGM-ST, 400 graph modeling samples are randomly selected. For example, in the RPaviaU-DPaviaC dataset, the graph modeling samples consist of five labeled samples from each of the seven classes and 365 remaining unlabeled samples.

C. Experiments on Hyperparameter Setting

The hyperparameter r determines the subspace dimensions of the spectral and spatial branches. Therefore, we investigate the impact of r on classification accuracies. Since AGM and AGM-ST differ only in the generation of pseudo labels, the experiments to determine the optimal value of hyperparameter

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TABLE VI

OA, AA, AND κ OF ABLATION STUDIES ON RPAVIAU-DPAVIAC, EHANGZHOU-RPAVIAHR, AND HOUSTON2018-HOUSTON2013 DATASETS

Loss				RPaviaU-DPaviaC			EHangzhou-RPaviaHR			Houston2018-Houston2013		
$\mathcal{L}_{\text{VAE}}$	$\mathcal{L}_{\text{DD}}$	$\mathcal{L}_{\text{B}}$	$\mathcal{L}_{\text{CE}}$	OA	AA	κ	OA	AA	κ	OA	AA	κ
	✓	✓	✓	0.8971	0.9094	0.8752	0.9417	0.9717	0.8707	0.8252	0.8344	0.7971
✓		✓	✓	0.8690	0.8935	0.8426	0.9803	0.9892	0.9519	0.8207	0.8284	0.7918
✓	✓		✓	0.5932	0.6517	0.5173	0.8011	0.8993	0.6286	0.6977	0.7093	0.6429
✓	✓	✓		0.8649	0.8841	0.8371	0.9396	0.9662	0.8629	0.8131	0.8180	0.7793
✓	✓	✓	✓	**0.9324**	**0.9397**	**0.9179**	**0.9949**	**0.9946**	**0.9869**	**0.8426**	**0.8471**	**0.8158**

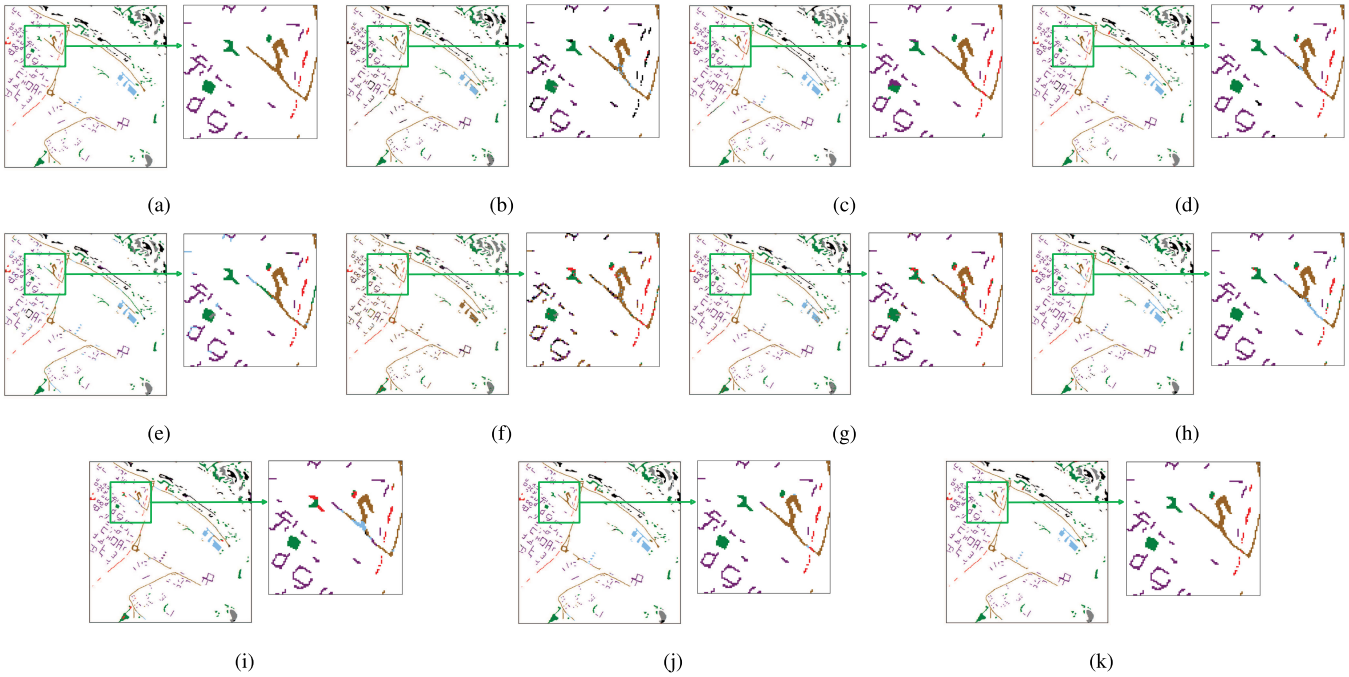


Fig. 9. Ground-truth and classification maps of RPaviaU-DPaviaC dataset. (a) Ground-truth map. (b) Classification map of SVM. (c) Classification map of HybridSN. (d) Classification map of TDSSFnet. (e) Classification map of 3-D-SRNet. (f) Classification map of DDNMF. (g) Classification map of CDVAE. (h) Classification map of Gia-CFSL. (i) Classification map of DCFSL. (j) Classification map of AGM. (k) Classification map of AGM-ST.

r are conducted exclusively on AGM. This optimal value of r is then utilized for both AGM and AGM-ST. We run experiments with $r \in \{5, 10, \ldots, 25\}$, and the results are visualized in Fig. 8. It can be observed that in all datasets, the accuracies (OA, AA, κ) exhibit an increasing trend as r increases within the range of [5, 20], and they remain unchanged after $r > 20$. In subsequent experiments, we select the highest accuracy achieved from the range $r \in \{5, 10, \ldots, 25\}$ as the final result for both AGM and AGM-ST.

D. Ablation Study

The quality of the generated graph weights in the first stage of AGM is significantly influenced by the choice of the loss function. To investigate this, we conduct an ablation study.

The total loss $\mathcal{L}_{\text{first}}$ in the first stage of AGM comprises four distinct loss functions: VAE loss $\mathcal{L}_{\text{VAE}}$, distribution discrepancy loss $\mathcal{L}_{\text{DD}}$, balancing loss $\mathcal{L}_{\text{B}}$, and cross-entropy loss $\mathcal{L}_{\text{CE}}$. In the ablation study, we systematically exclude one of these four losses at a time and evaluate the resulting classification accuracies.

Ablation studies are conducted on all three datasets, and the resulting classification accuracies were recorded, as shown in Table VI. Upon analyzing the results in Table VI, it is evident that removing any of the four losses leads to a decrease in classification accuracies. Therefore, all four losses in the first stage of AGM contribute positively to the overall performance.

TABLE VII

OA, AA, AND κ OF EACH ALGORITHM IN RPAVIAU-DPAVIAC DATASET

Method	OA	AA	κ
SVM	0.8297	0.8106	0.7937
HybridSN	0.7853	0.8065	0.7425
TDSSFnet	0.9211	0.9331	0.9044
3-D-SRNet	0.8460	0.8496	0.8135
DDNMF	0.6240	0.6250	0.5431
CDVAE	0.8701	0.8764	0.8460
Gia-CFSL	0.8781	0.8880	0.8524
DCFSL	0.8948	0.9053	0.8724
AGM	0.9324	0.9397	0.9179
AGM-ST	**0.9497**	**0.9510**	**0.9386**

E. Comparisons Among Different Algorithms

After discussing the hyperparameters in AGM-ST, a comparison among different methods is conducted. The results of all compared algorithms on the RPaviaU-DPaviaC dataset are presented in Table VII. Additionally, Fig. 9 displays the ground-truth and classification maps. It is important to note

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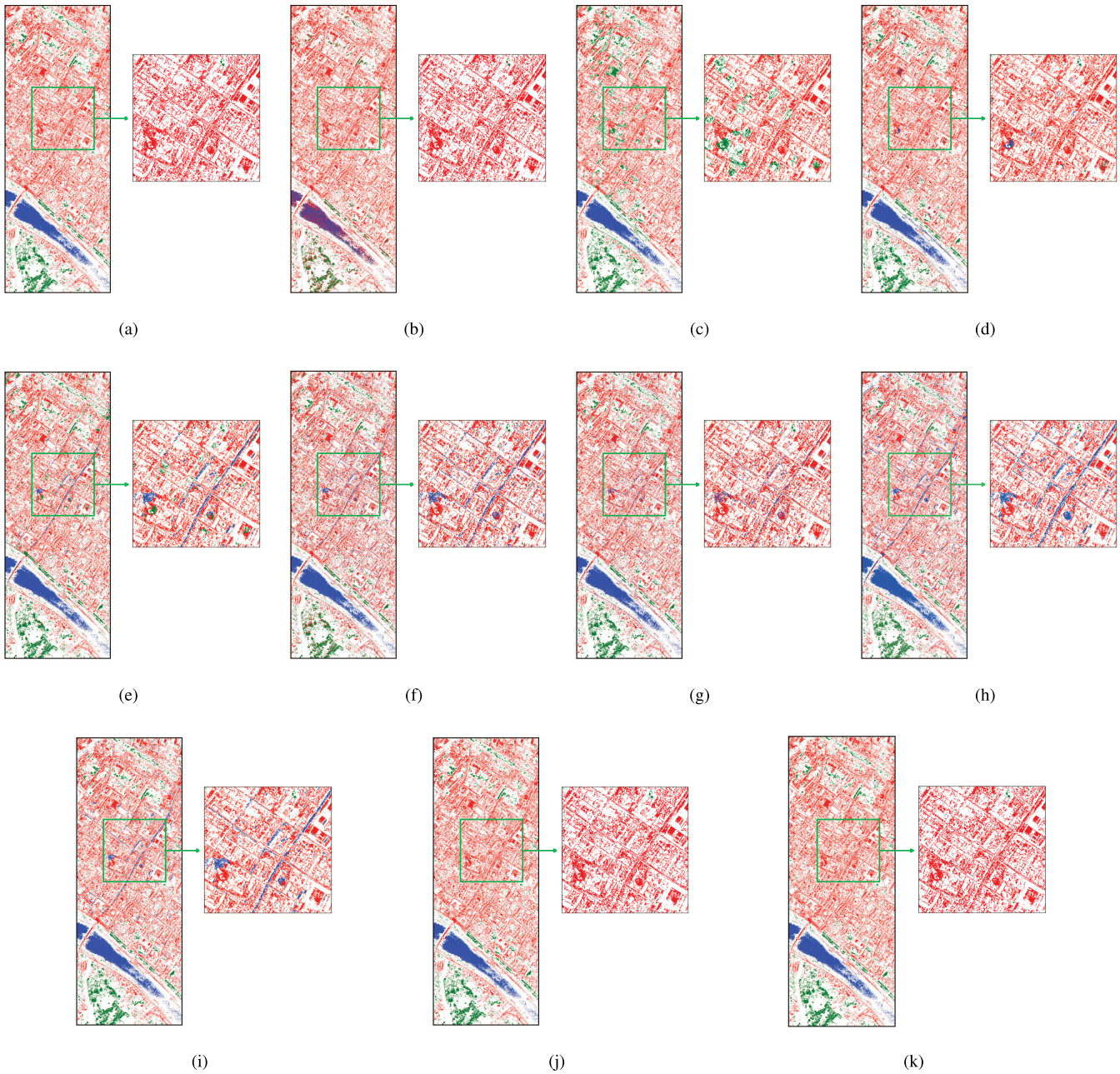


Fig. 10. Ground-truth and classification maps of EHangzhou-RPaviaHR dataset. (a) Ground-truth map. (b) Classification map of SVM. (c) Classification map of HybridSN. (d) Classification map of TDSSFnet. (e) Classification map of 3-D-SRNet. (f) Classification map of DDNMF. (g) Classification map of CDVAE. (h) Classification map of Gia-CFSL. (i) Classification map of DCFSL. (j) Classification map of AGM. (k) Classification map of AGM-ST.

that the accuracies are calculated using only the testing samples, while the classification maps are generated from both the training and testing samples. Furthermore, a specific segment with clear distinctions is selected for localized magnification in the ground-truth and classification maps. By analyzing the findings presented in Table VII, the following summary is drawn.

1) The deep learning algorithm HybridSN achieves lower accuracies compared to the shallow learning algorithm SVM, indicating the limitations of deep learning algorithms in extremely small-sample-size conditions. However, it is noteworthy that six cross-scene classification algorithms (3-D-SRNet, CDVAE, Gia-CFSL, DCFSL, AGM, and AGM-ST) achieve higher accuracies than the two single-scene classification algorithms (SVM and HybridSN). This observation highlights the effectiveness of transfer learning and underscores the significant contribution of cross-scene classification approaches.
2) DDNMF exhibits negative transfer in comparison to SVM, indicating that cross-scene classification is not always beneficial. It is important to note that only well-designed cross-scene classification models, such as 3-D-SRNet, CDVAE, Gia-CFSL, DCFSL, AGM, and

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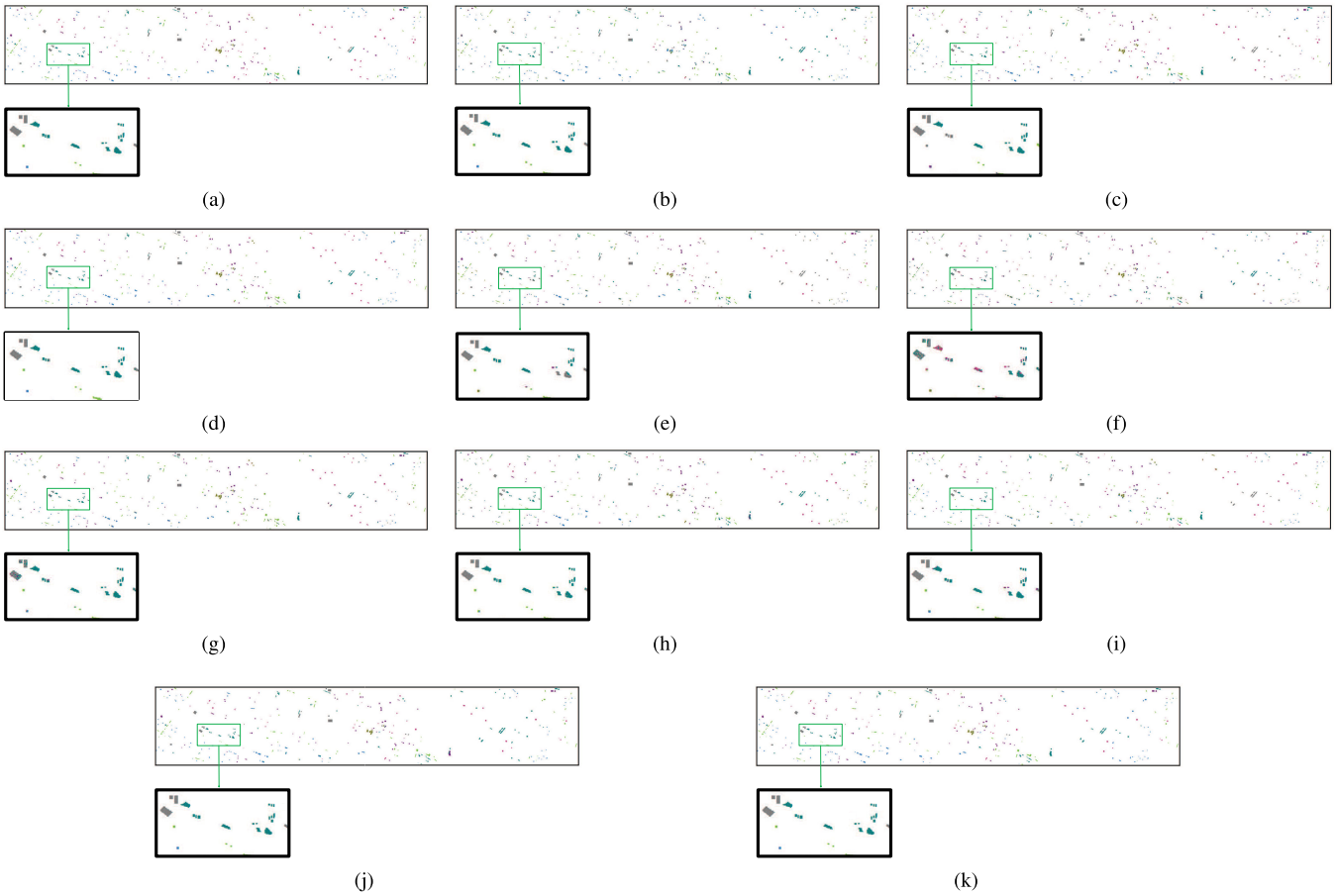


Fig. 11. Ground-truth and classification maps of Houston2018-Houston2013 dataset. (a) Ground-truth map. (b) Classification map of SVM. (c) Classification map of HybridSN. (d) Classification map of TDSSFnet. (e) Classification map of 3-D-SRNet. (f) Classification map of DDNMF. (g) Classification map of CDVAE. (h) Classification map of Gia-CFSL. (i) Classification map of DCFSL. (j) Classification map of AGM. (k) Classification map of AGM-ST.

AGM-ST, have the capability to enhance the classification performance in small-sample-size target scenes.

3) When comparing TDSSFnet with other single-scene classification methods, including SVM and HybridSN, it becomes evident that the spectral–spatial features extracted from the spectral and spatial branches of SSFB are more discriminative.
4) The accuracies achieved by AGM are higher than those of the other algorithms, with the exception of AGM-ST, which demonstrates the effectiveness of AGM in achieving superior classification performance.
5) The proposed AGM-ST achieves the best results due to the combined benefits of AGM and ST.

After analyzing the experimental results on the RPaviaU-DPaviaC dataset, we conducted additional experiments on the EHangzhou-RPaviaHR dataset. The OA, AA, and κ of each compared algorithm are presented in Table VIII. Furthermore, Fig. 10 showcases the ground-truth and classification maps of this dataset. The results obtained on the EHangzhou-RPaviaHR dataset exhibit similarities to those achieved on the RPaviaU-DPaviaC dataset. Consequently, we perform similar analyses based on the data presented in Table VIII.

1) In the EHangzhou-RPaviaHR dataset, HybridSN achieves slightly higher OA, AA, and κ compared to SVM. Moreover, the six cross-scene classification algorithms (3-D-SRNet, CDVAE, Gia-CFSL, DCFSL, AGM, and AGM-ST) outperform the two single-scene classification methods (SVM and HybridSN), demonstrating the performance gain achieved through cross-scene transfer learning.
2) Despite being a cross-scene classification algorithm, DDNMF may not be suitable when the target scene has an extremely limited number of labeled pixels.
3) As a single-scene classification method, TDSSFnet exhibits superior performance compared to several cross-scene methods, including 3-D-SRNet, DDNMF,

TABLE VIII
OA, AA, AND κ OF EACH ALGORITHM IN EHANGZHOU-RPAVIAHR DATASET

Method	OA	AA	κ
SVM	0.8554	0.7665	0.6577
HybridSN	0.8687	0.8682	0.7085
TDSSFnet	0.9625	0.9832	0.9137
3-D-SRNet	0.9192	0.9491	0.8212
DDNMF	0.7859	0.8631	0.5840
CDVAE	0.9416	0.9737	0.8711
Gia-CFSL	0.9317	0.9642	0.8463
DCFSL	0.9544	0.9765	0.8938
AGM	0.9949	0.9946	0.9869
AGM-ST	**0.9972**	**0.9971**	**0.9928**

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TABLE IX
OA, AA, AND κ OF EACH ALGORITHM IN HOUSTON2018-HOUSTON2013 DATASET

Method	OA	AA	κ
SVM	0.5715	0.5659	0.4918
HybridSN	0.6923	0.7144	0.6375
TDSSFnet	0.8324	0.8401	0.8056
3-D-SRNet	0.7838	0.7904	0.7451
DDNMF	0.6566	0.6837	0.5947
CDVAE	0.7627	0.7779	0.7201
Gia-CFSL	0.8028	0.8029	0.7674
DCFSL	0.8020	0.7953	0.7664
AGM	0.8426	0.8471	0.8158
AGM-ST	**0.8549**	**0.8614**	**0.8284**

CDVAE, Gia-CFSL, and DCFSL. This outcome highlights the effectiveness of the spectral and spatial branches in SSFB, as they aid in extracting representative and discriminative features.

4) AGM-ST demonstrates superior classification performance compared to AGM, thereby confirming the effectiveness of incorporating ST to generate pseudo labels during the AGM process.

For the Houston2018-Houston2013 dataset, we conducted the analysis using the same experimental settings as before. The OA, AA, and κ of each compared algorithm are presented in Table IX. Additionally, Fig. 11 displays the ground-truth and classification maps for this dataset. The results obtained on the Houston2018-Houston2013 dataset exhibit similarities to the previous two datasets. Therefore, we perform similar analyses based on the data presented in Table IX.

1) In the Houston2018-Houston2013 dataset, cross-scene classification algorithms generally outperform single-scene classification methods. This observation highlights the superiority of cross-scene classification algorithms in addressing small-sample-size problems.
2) Traditional machine learning algorithms (SVM and DDNMF) exhibit lower classification accuracies compared to deep learning algorithms (HybridSN, TDSSFnet, 3-D-SRNet, CDVAE, Gia-CFSL, DCFSL, AGM, and AGM-ST). This suggests that traditional machine learning algorithms are not well-suited for solving small-sample-size problems effectively.
3) Despite being a single-scene algorithm, TDSSFnet still performs admirably, surpassing most of the compared algorithms in terms of classification accuracy.
4) By comparing AGM and TDSSFnet, we can validate the effectiveness of transfer learning.
5) Overall, the proposed AGM-ST achieves the best performance among all the compared algorithms, demonstrating the effectiveness of combining AGM and ST in addressing small-sample-size problems.

V. CONCLUSION AND FUTURE WORK

In cross-scene classification, graph modeling is an effective technique for establishing correlations between samples from different scenes. However, the limited number of labeled samples in the target scene poses a challenge for many existing cross-scene modeling methods. These methods often fail to ensure both density and fault tolerance in the constructed graphs, resulting in weak relationships between samples or even negative transfer. To address this issue, we propose AGM-ST, an approach that strengthens sample relationships based on AGM and ST. AGM performs cross-scene graph modeling by leveraging the similarities between unlabeled samples through a learnable network. The adaptively generated graph is then utilized for training, facilitating further feature alignment and classification. Additionally, ST generates pseudo labels for selected samples with higher classification confidences, augmenting the semantic information and improving graph modeling. As a result, AGM-ST significantly improves the transfer learning performance for small-sample-size classification tasks. Experimental results on cross-scene HSIs have validated the effectiveness of AGM-ST.

In the current work, we incorporate graph information into the deep network using a graph loss function, which ensures that the generated features of the deep network exhibit cross-domain consistency. Recent research on graph convolutional networks (GCNs) has demonstrated the direct utilization of graphs in convolutional feature extraction. Moving forward, our future research will focus on exploring the potential of combining AGM with GCN, and investigating its applicability in cross-domain HSI classification.

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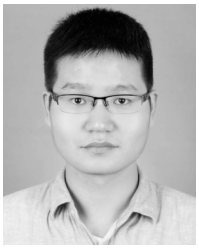
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